Mulgeta Mersha Cheru

Passport:EQ1791545 | **Work permit:**Ethiopian | **Date of birth:**19 Mar 1992 | **Place of birth:**North Shoa, Ethiopia | **Nationality:**Ethiopian | **Phone number:** (+251) 0910074638 (Mobile) | **Email address:**mulgeta_mersha@slu.edu.et | **Website:** http://www.mywebsite.com | **LinkedIn:**Your profile linkedin.com/in/mulgeta123 | **Facebook:** https://web.facebook.com/mulgeta.mersha | **X:** https://x.com/lalisamarsha |

Address: Charchil road, 245, Fiche, Ethiopia (Home) |

Address: Charchil road, 245, Fiche, Ethiopia (Work)

Additional Email address: lalisamarsha@gmail.com

ABOUT ME

I am a dedicated **structural engineer** holding both a BSc in civil engineering and an MSc in structural engineering. Possessing over six years of valuable experience as a university lecturer, effectively imparting core civil and structural engineering concepts to students. Served with distinction for four years as **Head of the Civil Engineering Department**, contributing to curriculum development and departmental strategic initiatives. Proven ability to design and deliver engaging courses, conduct impactful research, and contribute to departmental growth. Eager to leverage extensive technical expertise and pedagogical skills to foster student success and advance the institution's academic mission.

WORK EXPERIENCE

UNIVERSITY ENGINEERING LECTURER – ODA BULTUM UNIVERSITY – 21 Oct 2016 – 15 Oct 2017 – CHIRO, ETHIOPIA

Business or Sector: Education **Department:** Civil Engineering **Email:** mulgetamersha@obu.edu.et
Website: http://www.obu.edu.et

Instruction: I, an assistant lecturer, often lead tutorials, discussion sessions, and laboratory classes. Deliver lectures in subjects like structural analysis, Engineering Mechanics, Strength of materials, Theory of structure, Reinforced concrete, structural design, Steel and timber, bridge courses, etc.
Assessment: I assist in creating, administering, and grading assignments, quizzes, and exams. Create lecture notes, slides, assignments, and lab manuals
Mentorship: Assistant lecturers are often a primary point of contact for students, offering academic advice, guidance on course materials, and support with coursework. Break down complex engineering concepts into something students can actually use, not just memorize.

EDUCATIONAL RESEARCHER – JIMMA UNIVERSITY – 16 Oct 2017 – 30 Jan 2020 – JIMMA, ETHIOPIA

Business or Sector: Education **Department:** Civil Engineering department of **structural Engineering** Email: mulgetamersha@obu.edu.et **Website:** http://www.ju.edu.et

As educational researchers, we conduct research in the field of structural engineering. I strive to broaden the knowledge of how education processes, educational systems, and individuals (teachers and learners) work. They foresee areas of improvement and develop plans for the implementation of innovations.

UNIVERSITY ENGINEERING LECTURER – ODA BULTUM UNIVERSITY – 2 Feb 2020 – 9 Jun 2021 – CHIRO, ETHIOPIA

Business or Sector: Education **Department:** Civil Engineering Department **Email:** mulgetamersha@obu.edu.et
Website: http://www.obu.edu.et

1. Deliver lectures in subjects like structural analysis, Engineering Mechanics, Strength of materials, Theory ofstructure, Reinforced concrete, structural design, Steel and timber, bridge courses, etc.
2. Break down complex engineering concepts into something students can actually use, not just memorize. Designlesson plans, course outlines, and learning objectives.
3. Create lecture notes, slides, assignments, and lab manuals. Hold office hours and answer questions, as well as guide students on projects, internships, and career paths. Conduct research in areas like structural systems, materials, transportation, etc. Publish papers in journals, Apply for grants and funding

HEAD OF ACADEMIC DEPARTMENT – ODA BULTUM UNIVERSITY – 10 Jun 2021 – 30 Sep 2025 – CHIRO, ETHIOPIA

Business or Sector: Education **Department:** Civil Engineering **Address: Po box** 226, Chiro, Ethiopia
Email: mulgetamersha@obu.edu.et **Website:** https://www. obu.edu.et

Led the Civil Engineering Department, overseeing curriculum development, faculty coordination, and student academic advising.
Strategic Planning: Setting the department's goals and objectives in line with the institution's mission.
Curriculum Development: Overseeing the creation, revision, and assessment of academic programs and courses.
Faculty Development: Fostering professional growth among faculty members by supporting their teaching, research, and service activities.
Mentoring: Providing guidance and support to junior faculty and staff.
Personnel Management: Handling faculty and staff hiring, evaluations, and promotions.
Scheduling: Creating course schedules and assigning faculty teaching loads. Set long-term technical direction that align projects with organizational or national priorities
Introduce new technologies (e.g., BIM, automation, sustainable design) Handle design failures, delays, cost overruns and encourage research and development

CONSTRUCTION DESIGN ENGINEER – ODA BULTUM YNIVERSITY PROJECT OFFICE – 10 Sep 2021 – 30 Aug 2025 – CHIRO, ETHIOPIA

Business or Sector: Construction Department: Construction Project Office

1. **Contract administration and bid process:** Acting as a part of the evaluation committee in the construction project office. I was involved in all phases of the construction process, including bid document preparation, bid flotation, evaluation, and final contract award.
2. **On-site construction support:** I provide direct assistance for on-site construction work, specifically with landscaping site work and monitoring progress against the detailed construction schedule, identifying and proactively addressing potential issues and delays to keep the project on track.
3. Worked as a confirmatory and re-evaluation property valuation and compensation payments with the committee formed have acquired fruitful results.
4. **Verifying construction completion and final inspection**: I coordinate and ensure the successful completion of all required final inspections (e.g., building code, fire marshal) and commissioning tests for systems (e.g., HVAC, electrical, plumbing) to confirm functionality. Formalizing the date when the facility is sufficiently complete and ready for its intended use.

UNIVERSITY LECTURER IN ENGINEERING – SALALE UNIVERSITY – 17 Sep 2025 – Current – FICHE, ETHIOPIA

Business or Sector: Education **Department:** Civil Engineering **Address: Po box** 245, Fiche, Ethiopia **Email:** mulgeta_mersha@slu.edu.et
Website:https://www.slu.edu.et

Course Preparation: Designing course syllabi, creating lecture plans, and developing teaching materials such as presentations, handouts, and case studies.
Lecturing: Delivering lectures to students in a clear, engaging, and informative manner.
Assessment: Designing, administering, and grading exams, essays, and other assignments to evaluate student learning. Providing constructive feedback to help students improve.

Conducting Research: Planning and executing research projects, collecting and analyzing data, and interpreting results.
Publishing: Writing and submitting research papers to academic journals and presenting findings at conferences.
Mentorship: Mentoring junior staff and new lecturers.
Outreach: Representing the university at external events and engaging with the wider community.
Professional Development: Attending conferences, workshops, and training sessions to stay current in their field and improve their teaching and research skills.

EDUCATION AND TRAINING

16 OCT 2017 – 10 FEB 2020 Jimma, Ethiopia
STRUCTURAL ENGINEER JIMMA UNIVERSITY

Structural Analysis: This is a fundamental subject that teaches how to determine the effects of loads and forces on physical structures and their components.
Engineering Mechanics: This field includes topics like statics and dynamics. Statics deals with the analysis of structures at rest, while dynamics studies structures in motion or subject to time-dependent loads, such as those caused by earthquakes or wind.
Structural Design: This involves applying the principles of structural analysis and material science to create the detailed plans for structures.
Expertise in computer-aided design (CAD), analysis software like AutoCAD, Abaqus, ETABS, and SAP2000, as well as BIM, is crucial.
Teamwork: The ability to work effectively with others is key to successful project completion
The goal of this paper is to present the behavior and performance of a fully encased composite column subjected to a horizontal cyclic load using finite element simulation. A parametric study was conducted using the numerical model to investigate the influences of geometric properties of fully encased composite (FEC) columns. The geometric variables were the shape of structural steel, column slenderness ratio (L/D) / the ratio of the length of the column (L) to the depth of the column cross-section (D), and ties spacing ratio (s/D) / the ratio of transverse tie spacing (s) to the depth of the column cross-section (D). The nonlinear combined hardening and the concrete damage plasticity (CDP) model were used to define the properties of steel and concrete, respectively.
The finite element analysis results demonstrate that the profile of encased sections affects the performance of fully encased composite columns.

Website http://www.ju.edu.et **Field of study Structural Engineering Final grade** 3.91 **Level in EQF** level 8

Thesis The effects of different steel sections on the performance of encased composite columns under

cyclic lateral loads 13 JUN 2011 – 14 AUG 2016 Jimma, Ethiopia

CIVIL ENGINEER JIMMA UNIVERSITY

Problem-Solving and Critical Thinking
Project Management
Problem-Solving and Critical Thinking
Mathematics and Physics
Structural analysis and design of structural members
How to consider wind load while designing and analysis
Considering earthquake loads in structural design and analysis
Load combinations and their effects on the structure

Website http://www.ju.edu.et **Field of study: - Civil Engineering Final grade 3.74 Level** in

EQF level 8 **Thesis: -** Design and Analysis of basement plus ground plus six floor mixed use

building.

8 DEC 2024 – 15 DEC 2024 Addis Abab,
Ethiopia
SPECIFICATION AND BOQ PREPARATION TRAINING Ethiopian Association of Civil Engineers (EACA)

Understanding and Interpreting Project Documentation
Quantity Takeoff and Measurement
Specification Writing and Detailing
Cost Estimation and Pricing
Professional and Technical Skills

Website https://eacecivil.org/ **Field of study** Engineering, manufacturing and construction Level in EQF level 7

8 JUN 2021 – 21 JUN 2021 Arba Minch,
Ethiopia
CIVIL AND HYDRAULICS LABORATORY EQUIPMENT USE, APPLICATION AND MANAGEMENT TRANING
Arba-Minch University

Experimental Proficiency
Data Analysis
Safety and Management
Application of Principles

Website https://amu.edu.et/en/ **Field of study** Experimental Laboratory Training **Level in EQF** level 7

5 OCT 2020 – 1 OCT 2021 Harar, Ethiopia
HIGHER DIPLOMA PROGRAM IN TEACHING EDUCATION Haramay University

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Instructional Skills
Professionalism and Ethics
Educational Leadership

Website https://www.haramaya.edu.et/ **Field of study** HDP (Higher Diploma Education) **Level in EQF** level 7
8 OCT 2024 – 15 OCT 2024 Addis Ababa,
Ethiopia
RESILIENCE EARTHQUAKE SAFETY AND DESIGN FOR A SAFER FEATURE TRANING Ethiopian Association of Civil Engineers (EACA)

Structural and Non-Structural Safety
Risk and Vulnerability Assessment
Emergency Planning and Response
Design and Retrofitting Techniques

Field of study Education **Level in EQF**

level 7 1 OCT 2024 – 30 JUN 2025 Chiro,

Ethiopia

IELTS (INTERNATIONAL ENGILISH LANGUAGE TESTING SYSTEM) TRANING ODA BULTUM UNIVERSITY

It increases my day-to-day classroom English.
I increase the chance of winning different scholarships.

Website http://www.obu.edu.et **Field of study** Services **Level in EQF** level 7

LANGUAGE SKILLS

Mother tongue(s):
OROMO

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1		B2	B2	C1
AMAHARIC	C2	C2	C2	C2	C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Assist students in their learning, operate open source software, engineering principles, project management education administration, supervise educational staff performance, project management, curriculum standards, working with others, teaching and training.

HOBBIES AND INTERESTS

Listning Music, Reading Books, Swiming and watching Football

HONOURS AND AWARDS

12 JUL 2012

Best Pre engineering Student Award - Jimma University

The "Best Pre-engineering Student Award" is a recognition given to an outstanding student enrolled in a pre-engineering program. It's not a single, standardized award but rather a general term for various honors, scholarships, and prizes offered by the university.

NETWORKS AND MEMBERSHIPS

6 MAR 2021 – CURRENT Addis Ababa

Ethiopian Association of Civil Engineers (EACA),

The Ethiopian Association of Civil Engineers (EACE) is a professional organization that serves as a central hub for civil engineers in Ethiopia. The EACE aims to advance the civil engineering profession, disseminate knowledge, and contribute to the country's development and infrastructure.

PUBLICATIONS

2024

The effects of different steel sections on the performance of encased composite columns under cyclic lateral loads

The goal of this paper is to present the behavior and performance of a fully encased composite column subjected to a horizontal cyclic load using the finite element simulation. A parametric study was conducted using the numerical model to investigate the influences of geometric properties of fully encased composite (FEC) columns. The geometric variables were the shape of structural steel, column slenderness ratio (L/D) / the ratio of the length of the column (L) to the depth of the column cross-section (D), and ties spacing ratio (s/D) / the ratio of transverse tie spacing (s) to the depth of the column cross-section (D). The nonlinear combined hardening and the concrete damage plasticity (CDP) model were used to define the properties of steel and concrete, respectively.
The finite element analysis results demonstrate that the profile of encased sections affects the performance of fully encased composite columns. Encased Tube Section (ETS) columns had better performance than Encased Circular Section (ECS) and Encased H-Section (EHS) due to their high stiffness. Increasing the slenderness ratio of the column from 7.5 to 11.25 and 15 reduces the lateral resisting capacity by 6.74% and 15.83%, respectively. The tie spacing also has significant effects on the lateral resistance capacity of the composite column. The effect is more observed in the buckling effects: as the spacing of lateral ties increases, longitudinal reinforcement fails in buckling failure followed by crushing of concrete material before they reach their yielding stress.

Regasa yadeta Sembeta, Kefiyalew Zerfu, Mulgeta Mersha, Elmer C Agon

Authors: Mulgeta Mersha | **Journal Name**: Results in Engineering | **Volume, Issue and Pages**: Vol 4,Issue 2, Pages 103510 |
Publisher: Elsevier

REFERANCES

1. Tewodros Ghebrab, Ph.D., P.E.
Associate Professor of Civil, Environmental & Construction Engineering
Texas Tech University, Lubbock, TX, USA
Honorary Associate Professor of Jimma University
Tewodros.ghebrab@ttu.edu
(806) 834-3218

2. Kefiyalew Zerfu, (Dr.Eng)
Postdoctoral researcher at Yokohama National University
Yokohama National University, Graduate School of Urban Innovation
Yokohama, Japan
Honorary Lecturer of Jimma University
kefiyalewz@gmail.com